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Stress Testing Automated Trading Systems using a Quantitative Framework for Financial Regulatory Systems

Master's Thesis

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Abstract

The global financial sector is a highly regulated environment, and as AI models are adopted and deployed more widely within it, its regulation must become more precise about what it means for such a model to be robust and low-risk enough to deploy. Motivated by the requirements of Article 15 of the EU AI Act for high-risk AI systems, this thesis examines the robustness of an LSTM-based volatility prediction model under simulated financial stress, and, more fundamentally, the validity of the accuracy metric by which that robustness is judged. The goal is to build an end-to-end pipeline combining GJR-GARCH parameter estimation across four market periods, Monte Carlo generation of 10,000 stress paths across six scenarios, and an LSTM trained exclusively on calm 2010–2019 S&P 500 data, and then subject the resulting accuracy to a second-order analysis of its validity under stress. The research identifies and addresses a significant limitation in accuracy-based robustness assessment: a single accuracy figure can stay stable or shift for reasons unrelated to the model, because the raw metric tracks the composition of the evaluation labels rather than the model’s underlying skill. By constructing balanced accuracy from class-weighted rates and designing a lookback contamination test with a dependence-corrected bootstrap, the analysis establishes that raw accuracy is substantially more confounded with label composition than balanced accuracy ($r = -0.85$ versus $r = -0.64$; bootstrap difference -0.261 , 95% CI $[-0.441, -0.108]$). The thesis concludes that metric validity must be established before metric value is trusted, reframing robustness assessment under Article 15 as a governance question as much as a technical one and advancing this validity-first principle as a concrete path toward more trustworthy compliance assessment.

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List of Abbreviations

AI	Artificial Intelligence
ARCH	Autoregressive Conditional Heteroskedasticity
BalAcc	Balanced Accuracy
CI	Confidence Interval
COVID-19	Coronavirus Disease 2019
ESRB	European Systemic Risk Board
EU	European Union
FSB	Financial Stability Board
GARCH	Generalized Autoregressive Conditional Heteroskedasticity
GJR-GARCH	Glosten–Jagannathan–Runkle GARCH
GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
ns	Not Significant
ReLU	Rectified Linear Unit
RNN	Recurrent Neural Network
S&P 500	Standard & Poor’s 500 Index
TNR	True Negative Rate
TPR	True Positive Rate
VaR	Value-at-Risk

List of Symbols

P_t	Adjusted closing price at time t
r_t	Percent-scaled log return at time t
σ_t^2, σ_t	Conditional variance and standard deviation at time t
ω	Constant (baseline variance) term of the GJR-GARCH(1,1) model
α	ARCH coefficient (symmetric shock term)
γ	Leverage (asymmetry) coefficient
β	GARCH coefficient (volatility persistence)
ε_t	Innovation (shock) at time t
I_{t-1}	Negative-shock indicator (= 1 if $\varepsilon_{t-1} < 0$)
k	Shock multiplier scaling the baseline variance ω
ω_{adj}	Scaled baseline variance, $\omega_{\text{adj}} = \omega \times k$
$M(t)$	Exponential volatility multiplier (Non-linear Squeeze)
M_{max}	Asymptotic maximum of the squeeze multiplier
t_{shock}	Shock onset day of the Non-linear Squeeze
τ	Time constant of the exponential squeeze
y_t	Binary volatility label at day t (1 = HIGH, 0 = LOW)
X_t	60-day feature window of past returns (LSTM input)
L	Lookback window length (60 days)
w	Volatility averaging window (20 days)
P	Path matrix of Monte Carlo simulations
N	Number of Monte Carlo paths
T	Path length in trading days

$x_i, \tilde{x}_i$	Feature vector and its standardised form for path i
$\hat{\mu}_{\text{train}}, \hat{\sigma}_{\text{train}}$	Training-data mean and standard deviation (scaler)
f_{θ}	Trained LSTM model
$\hat{p}_i$	Predicted probability of HIGH volatility for path i
$\hat{y}_i$	Binary prediction for path i
Acc	Raw directional accuracy
BalAcc	Balanced accuracy
TP, TN, FP, FN	True/false positive and negative counts
TPR	True positive rate (accuracy on HIGH days)
TNR	True negative rate (accuracy on LOW days)
w_k, N_k	Class weight and sample count for class k
K	Number of classes ($K = 2$)
c	Contamination level (squeeze days in the feature window)
$\mathcal{C}$	Set of contamination levels $\{0, 3, 6, \dots, 51\}$
t^*	Shock onset day in the contamination protocol ($t^* = 200$)
$t_{\text{start}}, t_{\text{end}}$	Start and end indices of the feature window
π_{HIGH}	Proportion of HIGH labels in the evaluation set
r_{raw}	Correlation of π_{HIGH} with raw accuracy
r_{bal}	Correlation of π_{HIGH} with balanced accuracy
h	Cohen's h effect size
B	Number of bootstrap resamples

1 Introduction

To predict is to assume that the future will resemble the past in its recurring patterns, that is, in statistical terms, the process generating it remains stationary, and to extrapolate those dynamics forward. Quantifying risk and uncertainty has been a primary instrument of safety, from anticipating natural hazards and reading weather patterns to pricing risk in financial markets. Across millennia and domains, prediction has been humanity's continuous project, sorting signals, patterns, and behaviours into learnable regularities that reduce uncertainty. Yet history consistently demonstrates that predicting crisis is inherently different from other forms of prediction. With technological advances, we have entrusted this predictive work to machines but more importantly, the judgement of whether to trust them must remain ours.

The 2008 global financial crisis illustrates this failure precisely. Value-at-Risk (VaR), the primary tool major institutions used to measure and report market risk, failed to estimate the losses it was built to measure (Brunnermeier, 2009; Jorion, 2009). VaR reported stable risk levels until the day the system collapsed, because it was calibrated on historical data that did not contain the conditions that caused the crisis. The risk models behind it assumed the correlation between mortgage defaults across regional housing markets would stay low, as history had consistently shown; what that history could not anticipate was a national lending bubble that would drive house prices down across every region at once, pushing that correlation from low to strongly positive. The COVID-19 collapse of 2020 showcased a crisis pattern from a different origin. No volatility model, no VaR estimate, and no stress test had been calibrated for a health emergency threatening human life at scale. The crisis did not arrive through the financial system; it arrived entirely from outside, through channels no historical return series had ever contained.

This difficulty in anticipating crises persists within a far more capable, yet no less vulnerable, generation of systems. Artificial intelligence is now deeply integrated into financial markets, driving advances in volatility forecasting, risk pricing, and algorithmic trading at a scale and speed beyond earlier models. As these systems increasingly inform decisions with real consequences, the question shifts from the technical to the regulatory boards on how AI systems deployed in financial markets should be monitored, and on what evidence. The Financial Stability Board has identified AI model risk as a potential amplifier of systemic financial vulnerability (Financial Stability Board, 2024). An ESRB Advisory Scientific Committee report has recom-

mended revisiting prudential requirements for institutions deploying AI in systemically important functions (Cecchetti et al., 2025). In response, the EU AI Act will enter full application for high-risk AI systems. Under Article 15, these systems must demonstrate accuracy, robustness, cybersecurity, and perform consistently throughout their lifecycle (European Parliament and Council of the European Union, 2024). Robust oversight of such systems benefits from combining qualitative requirements with quantitative measurement, each capturing what the other cannot on its own. This thesis contributes on the quantitative side and works on two levels. Volatility estimation forms the risk-estimation layer of an automated trading system, since position limits, stop-losses, and VaR-based exposure all depend on it, which makes it the component most exposed to distributional shift under stress and the one this thesis examines. The first-order aim is to examine whether quantitative analysis can measure how these models perform under stress. The second-order aim, and the conceptual core of the work, is to test whether that accuracy metric remains valid under stress, so that any judgement of robustness rests on the model’s genuine skill rather than on the composition of the data it is evaluated on.

1.1 Research Question

This thesis investigates the following research question:

To what extent do changes in the accuracy of an LSTM-based volatility prediction model, under GJR-GARCH-generated stress scenarios, indicate a genuine loss of robustness relevant to Article 15 of the EU AI Act, and does the accuracy metric used to assess it remain valid under the conditions a stress test induces?

Three sub-questions follow:

1. How do GJR-GARCH parameters differ across normal and crisis periods, and what do these differences reveal about the volatility dynamics of distinct crisis regimes?
2. To what extent does LSTM performance change under progressively scaled GJR-GARCH stress scenarios?
3. To what extent can observed changes in accuracy be attributed to a genuine change in the model’s predictive skill, as opposed to shifts in label composition that alter the validity of the evaluation metric itself?

1.2 Contribution

This thesis makes three contributions.

Methodological. A reproducible stress-testing pipeline combining GJR-GARCH scenario generation with a lookback contamination protocol for testing the validity of the accuracy metrics used to evaluate the LSTM volatility prediction model under distributional shift, induced here by scaling volatility magnitude and by varying the label composition of the evaluation window.

Empirical. Evidence that raw accuracy tracks the label composition of the evaluation set more than the model’s underlying skill, showing that a single accuracy figure is an unreliable standalone measure of robustness under stress.

Policy-oriented. A validity-first governance approach for Article 15, in which an accuracy figure must pass a metric-validity check before it classifies risk, so a confounded metric escalates to human oversight rather than an automatic pass or fail.

1.3 Thesis Outline

Section 2 reviews the literature on GARCH volatility modelling, LSTM applications in finance, stress testing, and AI regulatory compliance. Section 3 sets out the methodology: data, GJR-GARCH estimation, Monte Carlo scenario generation, the LSTM architecture, and the stress-testing and lookback contamination protocols. Section 4 reports the empirical results, including the relationship between label composition and accuracy and its dependence-corrected significance. Section 5 discusses the findings, implications for Article 15 compliance, the limits of prediction in complex markets, and the role of human oversight. Section 6 concludes and outlines directions for future work.

2 Literature Review

2.1 GARCH Models for Volatility Forecasting

Volatility is a central concept in quantitative finance because it measures the variability and uncertainty in asset returns, yet it is not directly observable and must be inferred from historical patterns or suitable proxies. A core assumption of ordinary least squares is homoskedasticity, but financial returns are heteroskedastic: variance stays low during calm periods and rises sharply

during turbulent ones. This clustering of large price movements is a stylised fact of financial markets, documented by Mandelbrot (1963), who observed that large changes tend to be followed by large changes and small by small. Volatility therefore cannot be treated as constant, and requires models that capture how conditional variance responds to previous market movements.

The ARCH specification of Engle (1982) lets conditional variance depend on recent shocks, so volatility clustering emerges directly from the data. ARCH, however, needs many lagged terms to capture persistent shocks, and persistence is central to stress testing. Bollerslev (1986) generalised ARCH to GARCH, which captures both past shocks and past variance and so represents persistence more efficiently.

Standard GARCH nonetheless remains symmetric, because it depends on squared shocks, a -3% and a $+3\%$ shock raise future variance identically. This is a serious limitation for modelling stress, where adverse shocks affect future volatility differently from positive movements of equal size. Black (1976) first observed this leverage effect, and Glosten et al. (1993) formalised it by adding an asymmetry term directly to the variance equation. This thesis adopts the GJR-GARCH model because its asymmetry parameter γ enters the variance equation directly, in variance units, making it comparable across stress scenarios and crisis regimes, and allowing both persistence and asymmetric response to be built into the simulated scenarios.

2.2 Machine Learning for Financial Forecasting

Traditional GARCH models rely on a functional form specified in advance, with parameters estimated from the data. This suits volatility dynamics but limits their ability to model the nonlinear, nonstationary patterns present in financial data. Machine-learning models offer greater flexibility, learning patterns directly rather than assuming a predefined structure. Among these, recurrent architectures are designed for sequential data, but standard recurrent networks lose long-range dependencies as the error signal decays through the sequence. Hochreiter and Schmidhuber (1997) introduced the Long Short-Term Memory (LSTM) network to address this, using gated memory units that preserve information across long time lags, which makes the LSTM well suited to financial time-series forecasting.

Fischer and Krauss (2018, pp. 667–668) found that an LSTM achieved a directional accuracy (stock’s next-day return) of 54.3% for their primary $k = 10$ portfolio, declining toward roughly 51% as portfolio size increased;

although the margins are narrow, the LSTM was statistically superior to random forests, deep neural networks, and logistic regression. More recently, Yachou et al. (2026, pp. 17, 20) reported that LSTM performance deteriorated markedly during COVID-19, with R^2 falling from 0.95 to 0.89, a degradation they attributed to the model’s reliance on historical price data alone, without exogenous factors. This points to a broader limitation of purely data-driven models where they learn relationships specific to their training period and can degrade when market conditions move outside it, a problem examined more closely in Section 2.4. This thesis uses both models to create one end-to-end pipeline, where the GARCH process builds the stress scenarios and the LSTM is then tested on what it produces.

2.3 Financial Stress Testing and Monte Carlo Scenario Generation

Stress testing complements statistical forecasting because strong performance in calm periods says very little about how a model holds up in a crisis. It became central after 2008, when historically calibrated models such as Value-at-Risk underestimated the chance of many large moves arriving at once (Jorion, 2009; Basel Committee on Banking Supervision, 2009). Generating such scenarios calls for Monte Carlo simulation, which builds large numbers of hypothetical paths by sampling from a specified stochastic process (Glasserman, 2003). Historical simulation can only resample returns that have already occurred; a parametric approach instead draws paths from an estimated process, so the severity of a scenario can be pushed beyond anything in the historical record. This is precisely the capability that the synthetic-extreme and non-linear-squeeze regimes in this thesis require.

2.4 AI Regulatory Compliance and Evaluation Under Distribution Shift

The increasing adoption of AI in the financial domain has made model reliability a regulatory requirement rather than merely an engineering preference. The EU AI Act (European Parliament and Council of the European Union, 2024), the first horizontal legal framework for AI, sets binding requirements for high-risk AI systems, including those used in high-risk financial applications. These fall under Article 15, which requires an appropriate level of accuracy, robustness, and cybersecurity, performed consistently across the

system’s lifecycle, and Article 15(2) calls for benchmarks and measurement methodologies against which those properties can be assessed. The concern is echoed at the systemic level: the Financial Stability Board (2024) treats AI model risk as a potential amplifier of correlated failures, and an ESRB Advisory Scientific Committee report recommends revisiting prudential requirements for institutions deploying AI in systemically important functions (Cecchetti et al., 2025).

What this leaves open is how a model, and the metric judging it, should be validated under the very conditions where robustness must hold. Fan et al. (2023) formalise the distribution-shift problem for time-series forecasting, showing that performance degrades as inputs drift from the training distribution. For crisis evaluation this is not an incidental risk but its defining feature, since a stress test exists precisely to move a model away from the regime it learned. The concern this thesis builds on runs deeper: the shift can change the meaning of the metric used to measure it. Ye et al. (2024) survey spurious correlations, where evaluation comes to rest on associations that hold in one distribution but not another; a related effect is the label base rate. This thesis therefore reports two metrics: raw accuracy, computed on each scenario’s actual label mix and moving whenever that mix shifts, and balanced accuracy, which weights the true-positive and true-negative rates equally and is far less sensitive to it. The gap, then, lies not in the standard but in its verification: regulators define the standards high-risk AI must meet, while quantitative research supplies the methods that test whether they hold under stress. This thesis contributes to the second, arguing that sound compliance rests on policy, quantitative methodology, and human judgement together rather than on any single measure.

3 Methodology

This chapter presents the methodological framework used to stress test an AI-based volatility prediction model under extreme simulated scenarios and to examine whether the metrics used to evaluate it remain valid under those conditions. The framework consists of five components: (1) data collection and preprocessing, (2) GJR-GARCH parameter estimation for volatility dynamics, (3) Monte Carlo stress scenario generation, (4) LSTM model development for volatility prediction, and (5) stress testing evaluation, including the look-back contamination test.

3.1 Data Source and Processing

This thesis uses daily adjusted closing prices of the S&P 500 index (ticker ^GSPC) from Yahoo Finance, covering January 1997 to December 2022. The data were downloaded with the `yfinance` Python library and cached locally in Parquet format for reproducibility. This period includes the 2008 Global Financial Crisis and the 2020 COVID-19 pandemic, whose severities set the reference points for the stress scenarios.

3.2 Data Loading Architecture

A custom `SP500DataLoader` class handles data loading and caches each pull in Parquet format, so a given period is downloaded once and reused across runs. The loader takes a start and end year.

2008 crisis structure: `start_year = 1997, end_year = 2009`. The calm pre-crisis phase runs from 1997 to 2007, and the crisis phase (2007–2009) captures the 2008 drawdown.

COVID-19 structure: `start_year = 2005, end_year = 2022`. The 2005–2009 data serve only as GARCH warm-up; the calm training period spans 2010–2019, and the COVID crisis period spans 2020–2021.

3.3 Return Calculation

Logarithmic returns were computed as:

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right) \times 100 \quad (1)$$

where P_t is the adjusted closing price at time t . Log returns are continuously compounded and time-additive, the standard input for volatility models such as GARCH. The $\times 100$ scaling expresses them in percentage points and is applied consistently across GARCH estimation, Monte Carlo generation, and LSTM training, so that all components operate on the same scale and the optimisation stays numerically stable.

3.4 GARCH Model Specification

GARCH (Bollerslev, 1986) models volatility clustering, the tendency for high-volatility days to follow high-volatility days and calm to follow calm (Mandelbrot, 1963), by making the conditional variance depend on past shocks and past variance.

This framework uses the GJR-GARCH model, which extends GARCH with an asymmetry term (Glosten et al., 1993). Standard GARCH is symmetric, so positive and negative shocks of equal size have the same effect on future volatility. Empirically, negative shocks raise future volatility more than positive ones, an effect known as the leverage effect (Black, 1976).

The GJR-GARCH(1,1) model is specified as:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma I_{t-1} \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (2)$$

where $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$ (negative shock) and 0 otherwise. The parameter γ captures the leverage effect: negative shocks increase future volatility by an additional γ over the symmetric impact α , while positive shocks contribute only α .

Table 1 presents the estimated GJR-GARCH parameters for all four periods.

Period	ω	α	γ	β	Persistence
Pre-2008 (1997–2007)	0.016064	0.0000	0.1359	0.9204	0.9884
2008 Crisis (2007–2009)	0.028434	0.0000	0.1301	0.9199	0.9849
Pre-COVID (2010–2019)	0.035730	0.0000	0.2793	0.8161	0.9558
COVID Crisis (2020–2021)	0.114953	0.1762	0.3328	0.6238	0.9664

Table 1: GJR-GARCH(1,1) parameter estimates by period (percent-scaled log returns).

3.5 Monte Carlo Stress Scenario Generation

Monte Carlo simulation generates synthetic paths from the GJR-GARCH process defined above, using the estimated pre-COVID (2010–2019) parameters as the baseline “Normal Market” dynamics (Glasserman, 2003). The pre-COVID period is the baseline because it is the period on which the LSTM is trained, so each stress scenario is a controlled deviation from the model’s own training structure, on which the trained model is then evaluated. For each scenario, 10,000 independent paths of 252 trading days (one calendar year) were

generated, giving standard errors below 0.5% for the accuracy estimates.

The shock multiplier k scales the baseline variance parameter ω as $\omega_{\text{adj}} = \omega \times k$, while the other parameters (α, γ, β) are held constant to isolate the effect of volatility magnitude. Because the dynamics (α, γ, β) are held fixed, the effect of this scaling on the process has a closed form. The unconditional variance of the GJR-GARCH process is

$$\bar{\sigma}^2 = \frac{\omega}{1 - (\alpha + \beta + \frac{1}{2}\gamma)}, \quad (3)$$

so scaling ω by k leaves the persistence term unchanged and scales the long-run variance linearly, $\bar{\sigma}_{\text{adj}}^2 = k \bar{\sigma}^2$, which corresponds to scaling the long-run volatility by $\sqrt{k}$ (full derivation in Appendix A.1). The multiplier k therefore controls magnitude alone, while the GARCH persistence, leverage asymmetry, and ARCH sensitivity remain those of the pre-COVID regime. The multipliers are relative to the pre-COVID baseline and mark stress intensity, so a label such as “2008 crisis magnitude” denotes a stress level rather than the real volatility of the 2008 crisis. The Structural Break scenario introduces a mid-path step in the volatility level: ω is held at the baseline until day 180 and then steps up to $3.0 \times$ the baseline for the remainder of the path, with the dynamics (α, γ, β) unchanged across the break. The Non-linear Squeeze scenario applies an exponentially increasing multiplier after day 200:

$$M(t) = 1 + (M_{\text{max}} - 1) \left(1 - e^{-(t - t_{\text{shock}})/\tau} \right),$$

with $M_{\text{max}} = 5.0$, $t_{\text{shock}} = 200$, and $\tau = 10$. The multiplier is applied as a deterministic overlay on the simulated volatility path, scaling σ_t after the GJR-GARCH recursion rather than entering the variance recursion, so that the escalating stress does not compound explosively through the ARCH terms. Representative single-path simulations of the Structural Break and Non-linear Squeeze scenarios are shown in Figure 4 (Appendix A.2).

Scenario	Shock Multiplier	Description
Normal	$1.0\times$	Baseline conditions
Covid Crisis Magnitude	$1.8\times$	Elevated volatility
2008 crisis Magnitude	$2.5\times$	Severe volatility
Synthetic Extreme	$3.0\times$	Beyond historical
Structural Break	$1.0\times \rightarrow 3.0\times$ (day 180)	Mid-path regime switch
Non-linear Squeeze	Exponential (day 200)	Accelerating panic

Table 2: Monte Carlo stress scenario definitions.

The COVID regime differs from the pre-COVID baseline in its dynamics as well as its variance level. Scaling ω alone is therefore a deliberate choice, isolating the effect of the variance level from a simultaneous shift in dynamics.

3.6 LSTM Model for Volatility Prediction

3.6.1 Model Selection Rationale

The central question of this thesis is whether an AI-based volatility prediction model can meet the robustness required of high-risk financial systems, and whether the metric used to assess that robustness remains valid under stress. Several architectures are used for financial time-series forecasting, including random forests, LSTMs, and transformers. Preliminary experiments with a Random Forest, a Directional LSTM, and an Attention LSTM (Breiman, 2001; Hochreiter and Schmidhuber, 1997) led to the regularised LSTM being selected for the main analysis. The regularised LSTM, following the design of Yachou et al. (2026, pp. 7–8), was chosen because it captures the temporal structure of the return sequence, while its dropout and learning-rate scheduling control overfitting on the limited calm-period sample, and its simplicity suits a thesis aimed at testing metric validity rather than maximising accuracy.

The main analysis uses LSTM-based volatility prediction, forecasting whether next-day volatility will be high or low relative to a recent baseline. Volatility is more predictable than return direction because it clusters. The LSTM suits this task and has performed well in financial sequence forecasting (Fischer and Krauss, 2018). This thesis adapts the LSTM architecture of Yachou et al. (2026, pp. 7–8), used there for asset-price prediction, to the distinct task of volatility prediction.

3.6.2 LSTM Task Formulation

The LSTM predicts whether next-day volatility will be HIGH or LOW relative to the recent 20-day average. For a feature window ending on day $t - 1$, the binary classification target for day t is defined as:

$$y_t = \mathbf{1} \left(|r_t| > \frac{1}{20} \sum_{i=1}^{20} |r_{t-i}| \right) \quad (4)$$

where:

- $|r_t|$ is the absolute return (volatility proxy) on the target day t , which lies immediately after the feature window,
- the denominator is the average absolute return over the 20 trading days preceding the target day,
- $y_t = 1$ indicates HIGH volatility (above average),
- $y_t = 0$ indicates LOW volatility (below average).

Input features are a 60-day rolling window of past returns ending on the day before the target:

$$X_t = [r_{t-60}, r_{t-59}, \dots, r_{t-1}] \quad (5)$$

The target day t is deliberately excluded from the feature window to prevent look-ahead bias, and the same convention is applied at training and at evaluation (Section 3.7). The 60-day lookback (approximately one quarter) was chosen from preliminary experimentation and matches the 60-day window found optimal in the sensitivity analysis of Yachou et al. (2026, pp. 5, 17).

The dataset is imbalanced, with approximately 59.6% LOW and 40.4% HIGH labels. Balanced class weights are applied during training so that misclassifying the minority HIGH-volatility class carries a higher penalty; without them the model tends to predict LOW almost always, scoring high naive accuracy while missing the HIGH-volatility events that stress testing exists to catch. For the same reason, results report balanced accuracy alongside raw directional accuracy, since raw accuracy on an imbalanced set can be dominated by the majority class (Section 3.7).

3.6.3 LSTM Model Architecture

The LSTM model architecture consists of:

Layer	Details	Parameters
LSTM	32 units, return_sequences=False	4,352
Dropout	Rate = 0.2	0
Dense	16 units, ReLU activation	528
Dense	1 unit, Sigmoid activation	17
Total		4,897

Table 3: LSTM architecture for volatility prediction.

The architecture uses a single LSTM layer with 32 units to compress the 60-day return sequence into one hidden state, like condensing a long story into 32 summary notes. A dropout layer (rate 0.2) reduces overfitting by forcing the model to rely on multiple cues rather than memorising noise. A dense layer with 16 units and ReLU activation extracts non-linear patterns, and a final dense layer with sigmoid activation outputs the probability of HIGH volatility. This architecture follows Yachou et al. (2026, pp. 7–8), with the sigmoid output and binary cross-entropy replacing their regression head for classification and balanced class weights added for the label imbalance. With only 4,897 trainable parameters, the model stays small relative to its task.

The learning rate is set to 0.0005, following Yachou et al. (2026, pp. 5, 17) and below the Adam default of 0.001. Like smaller steps on an icy surface, this slower optimisation gives steadier, more stable convergence and helps generalisation. In-domain validation performance is reported in Section 4.

3.6.4 Training Configuration

The model was trained exclusively on normal market conditions (2010–2019). Key hyperparameters are summarised below.

Hyperparameter	Value
Training period	2010–2019 only
Lookback window	60 days
Train/validation split	80/20 (chronological)
Loss function	Binary cross-entropy
Optimiser	Adam (lr = 0.0005)
Batch size	32
Maximum epochs	100
Early stopping patience	10
Class weights	Balanced (HIGH 40.4% / LOW 59.6%)
ReduceLROnPlateau	patience = 5, factor = 0.5
Random seed	42 (fixed for reproducibility)

Table 4: LSTM training configuration.

3.7 Stress Testing Protocol

For each Monte Carlo scenario comprising $N = 10,000$ simulated paths of length $T = 252$ trading days, the trained LSTM model was evaluated using a standardised five-step protocol. Let $\mathbf{P} \in \mathbb{R}^{N \times T}$ denote the path matrix, $L = 60$ the lookback window, and $w = 20$ the volatility averaging window.

1. **Feature extraction.** From each path i , the 60-day feature window immediately preceding the label target day is extracted as:

$$\mathbf{x}_i = \mathbf{P}_{i, -(L+1):-1} \in \mathbb{R}^L \quad (6)$$

The window is indexed as $[-(L+1) : -1]$ rather than $[-L :]$ to exclude the label target day from the feature sequence, preventing look-ahead bias. This ensures the model never observes the return it is attempting to predict, and matches the feature convention used during training.

2. **Feature standardisation.** Each feature vector is standardised with a `StandardScaler` whose mean and standard deviation are estimated once, from the training data $\mathbf{X}_{\text{train}}$ (2010–2019), and then applied unchanged to every stress-scenario evaluation:

$$\tilde{x}_{i,t} = \frac{x_{i,t} - \hat{\mu}_{\text{train}}}{\hat{\sigma}_{\text{train}}} \quad (7)$$

The training statistics act as a fixed reference for what counts as normal. Measured against this calm baseline, a crisis-period return still stands out as extreme, which is exactly what the stress test needs to see. Fitting a new scaler on each evaluation set would instead measure crisis returns against other crisis returns, making the stress period look ordinary and hiding the distributional shift the test is designed to expose.

3. **Volatility label construction.** The binary volatility label for each path i is defined as:

$$y_i = \mathbf{1} \left(|r_{i,T}| > \frac{1}{w} \sum_{j=1}^w |r_{i,T-j}| \right) \quad (8)$$

where $r_{i,T}$ is the return on the label target day (the final day of the path) and the summation runs from $j = 1$ to $j = w$, so exactly $w = 20$ observations enter the recent volatility baseline. In vectorised form across all N paths:

$$\bar{v}_i = \text{mean}(|\mathbf{P}_{i, -(w+1):-1}|, \text{axis} = 1), \quad y_i = \mathbf{1}(|\mathbf{P}_{i, -1}| > \bar{v}_i) \quad (9)$$

4. **Prediction.** The scaled feature vector is reshaped to the LSTM input format $(N, L, 1)$ and passed through the trained model f_θ :

$$\hat{p}_i = f_\theta(\tilde{\mathbf{x}}_i) \in [0, 1] \quad (10)$$

where $\hat{p}_i$ is the predicted probability that next-day volatility will be HIGH. Predictions are generated in batches of 500 paths for memory efficiency.

5. **Accuracy calculation.** A binary prediction is obtained by thresholding at 0.5:

$$\hat{y}_i = \mathbf{1}(\hat{p}_i > 0.5) \quad (11)$$

Raw directional accuracy across all N paths is:

$$\text{Acc} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(\hat{y}_i = y_i) \quad (12)$$

Because the label distribution differs across scenarios, balanced accuracy is also reported as the mean of the true-positive and true-negative rates:

$$\text{BalAcc} = \frac{1}{2} \left(\frac{\text{TP}}{\text{TP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP}} \right), \quad (13)$$

which is independent of shifts in the proportion of HIGH labels and therefore isolates genuine discriminative skill from changes in class balance (Brodersen et al., 2010, p. 3121).

Class weight balancing. The balanced class weights introduced in Section 3.6.2 are computed as:

$$w_k = \frac{N_{\text{train}}}{K \cdot N_k} \quad (14)$$

where N_{train} is the total number of training samples, $K = 2$ the number of classes, and N_k the number of samples in class k . This gives weights of approximately 0.839 (LOW) and 1.237 (HIGH), penalising misclassification of the minority HIGH-volatility class more heavily.

3.8 Lookback Contamination Protocol

A lookback contamination test (window shifting) examines whether the model's measured performance on the Non-linear Squeeze scenario reflects genuine predictive skill or is an artefact of the evaluation metric. The Non-linear Squeeze is used because its volatility multiplier increases smoothly and monotonically after the shock onset, so the proportion of squeeze-period (HIGH) labels changes gradually as the feature window moves, producing the range of label proportions the test requires. The test slides the 60-day feature window to increasing depths into the scenario, varying the proportion of squeeze-period returns visible to the model while holding all other parameters constant.

Formal Definition

Let $\mathbf{P} \in \mathbb{R}^{N \times T}$ denote the matrix of $N = 10,000$ Monte Carlo paths, each of length $T = 252$ trading days. Let $t^* = 200$ denote the shock onset day of the Non-linear Squeeze scenario, and let $L = 60$ denote the fixed lookback window length.

For each contamination level $c \in \mathcal{C} = \{0, 3, 6, \dots, 51\}$ (18 evenly spaced levels; the maximum, $c = 51$, is the deepest position for which a label day remains within the path), the evaluation protocol proceeds as follows:

Step 1: Window positioning. The end and start indices of the feature window are defined as:

$$t_{\text{end}}(c) = t^* + c \quad (15)$$

$$t_{\text{start}}(c) = t_{\text{end}}(c) - L \quad (16)$$

so the window always spans $L = 60$ days, of which c fall inside the squeeze and $L - c$ precede it. As c grows this split shifts, from a window lying entirely before the shock at $c = 0$ to one sitting almost fully inside the squeeze at the deepest level, while the target day $t_{\text{end}}(c)$ is the day immediately after the window and never enters the input. Figure 1 shows all 18 levels: each step in c moves the 60-day window three days deeper, so the windows fan out from fully calm to almost fully squeezed. Calm pre-squeeze days appear in blue and squeeze days in red, shaded by the multiplier $M(t)$ from light at the onset to dark toward $M_{\text{max}} = 5$; the target day is marked in black, and the top panel plots the squeeze multiplier, which begins at day $t^* = 200$.

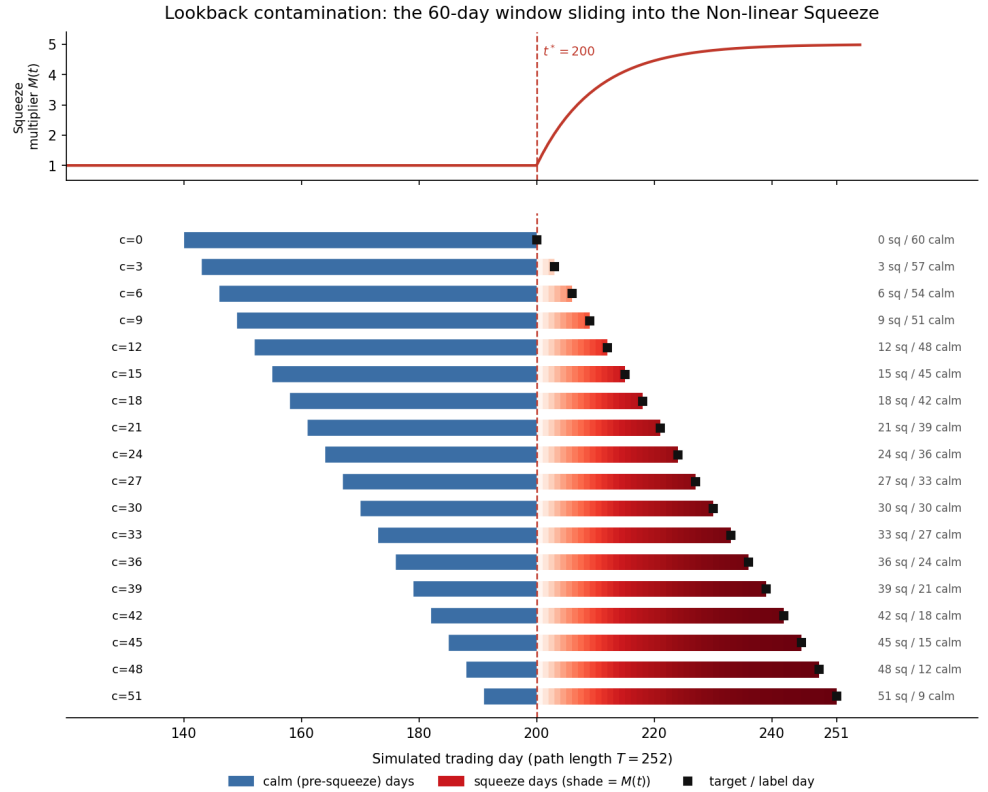


Figure 1: Construction of the 18 contamination levels.

Step 2: Feature extraction. For each path $i \in \{1, \dots, N\}$, the feature vector is extracted as the subsequence of the L days preceding the target day $t_{\text{end}}(c)$:

$$\mathbf{x}_i(c) = \mathbf{P}_{i, t_{\text{start}}(c):t_{\text{end}}(c)} \in \mathbb{R}^L \quad (17)$$

so that the target day $t_{\text{end}}(c)$ is excluded from the features, consistent with the stress testing protocol.

Step 3: Feature scaling. Each feature vector is standardised using the StandardScaler fitted exclusively on the training data $\mathbf{X}_{\text{train}}$ (2010–2019), consistent with the stress testing protocol of Section 3.7:

$$\tilde{\mathbf{x}}_i(c) = \frac{\mathbf{x}_i(c) - \hat{\boldsymbol{\mu}}_{\text{train}}}{\hat{\boldsymbol{\sigma}}_{\text{train}}} \quad (18)$$

Step 4: Label construction. The binary volatility label for each path i at contamination level c is defined as:

$$y_i(c) = \mathbf{1} \left(|r_{i, t_{\text{end}}(c)}| > \frac{1}{w} \sum_{j=1}^w |r_{i, t_{\text{end}}(c)-j}| \right) \quad (19)$$

where $w = 20$ is the rolling volatility window and $r_{i,t}$ denotes the log return of path i at day t . A label of $y_i(c) = 1$ indicates HIGH volatility relative to the recent 20-day average; $y_i(c) = 0$ indicates LOW volatility.

Step 5: Prediction and accuracy. The trained model f_{θ} maps each scaled feature sequence to a predicted probability of HIGH volatility, and a binary prediction is obtained by thresholding this probability at 0.5. At each contamination level c , both the raw directional accuracy $\text{Acc}(c)$ and the balanced accuracy $\text{BalAcc}(c)$ are recorded.

Step 6: Label distribution measurement. The proportion of HIGH volatility labels at each contamination level is recorded as:

$$\pi_{\text{HIGH}}(c) = \frac{1}{N} \sum_{i=1}^N y_i(c) \quad (20)$$

Step 7: Metric validity assessment. To test whether the evaluation metric tracks label composition rather than model skill, the Pearson correlation between the HIGH-label proportion and each accuracy measure is computed

across all contamination levels:

$$r_{\text{raw}} = \text{Corr}(\{\pi_{\text{HIGH}}(c)\}, \{\text{Acc}(c)\}), \quad r_{\text{bal}} = \text{Corr}(\{\pi_{\text{HIGH}}(c)\}, \{\text{BalAcc}(c)\}). \quad (21)$$

A raw correlation substantially stronger than the balanced correlation indicates that raw accuracy is confounded with the label distribution under distributional shift. Balanced accuracy, which weights the two classes equally, is far less sensitive. Because the 18 levels are overlapping windows over the same paths, they are not independent and the nominal Pearson p -values are anti-conservative. The difference $r_{\text{raw}} - r_{\text{bal}}$ is therefore assessed by a path-level bootstrap (Efron and Tibshirani, 1993): the 10,000 paths are resampled with replacement, the 18-level protocol is recomputed on each resample, and the resulting distribution of the difference gives a 95% percentile confidence interval. An overview of the complete methodology pipeline is provided as a flowchart in Appendix A.6, and Algorithm 1 (Appendix A.7) summarises the full contamination procedure as pseudocode.

4 Results

4.1 GJR-GARCH Parameter Estimates

Table 1 presents the estimated GJR-GARCH(1,1) parameters across all four periods. The leverage effect (γ) is positive in every period and statistically significant in three of the four: the pre-2008, 2008 crisis, and pre-COVID periods ($p < 0.05$), confirming the leverage effect (Black, 1976; Glosten et al., 1993). The COVID crisis estimate is the largest in magnitude ($\gamma = 0.3328$) but does not reach significance ($p = 0.134$), which reflects the short 505-day sample dominated by the March 2020 volatility spike.

The COVID crisis also stands out as a structurally distinct regime. The baseline variance ω rises roughly sevenfold, from 0.016064 in the pre-2008 period to 0.114953 during the COVID crisis; the GARCH coefficient β falls from 0.9204 to 0.6238, and the symmetric ARCH term α becomes non-zero (0.1762) only in this period. These differences indicate that the COVID regime departs from the 2008 crisis in dynamics, not only in variance level. All four periods satisfy the stationarity condition $\alpha + \beta + \gamma/2 < 1$, confirming that the estimated processes are mean-reverting, while the near-unity persistence in the earlier periods indicates that volatility shocks dissipate slowly.

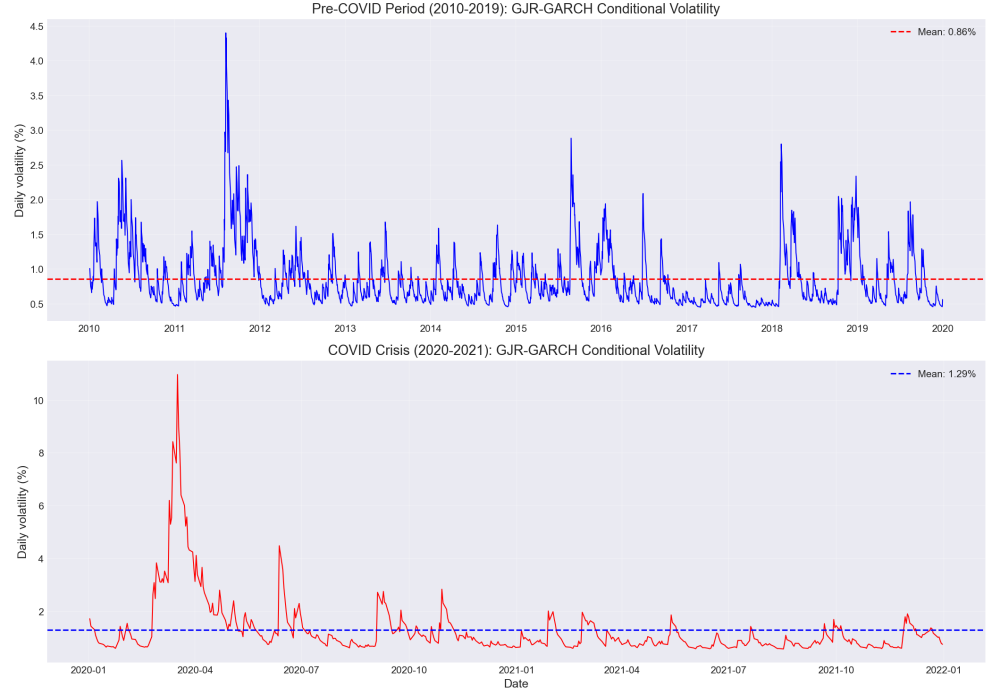


Figure 2: Conditional Volatility: Pre-COVID and COVID Crisis Periods

4.2 Model Validation Performance

The regularised LSTM achieved a validation accuracy of 66.80%, a balanced accuracy of 64.76%, and a validation loss of 0.6346 on the held-out portion of the 2010–2019 training data (the final 20% of the chronological split). This exceeds the naive baseline of 59.6%, obtained by always predicting the majority (LOW) class. Early stopping was triggered at epoch 31, restoring the best weights from epoch 21.

Because a majority-class predictor scores only 50% on balanced accuracy, the balanced accuracy of 64.76% shows that the model detects HIGH volatility events and does not simply predict the majority class. This establishes a meaningful in-domain skill level against which the model’s behaviour under stress can be assessed. The training and validation curves (Figure 5, Appendix A.3) show stable convergence, with the best weights restored at epoch 21.

4.3 Stress Test Results

Table 5 presents the directional accuracy of the regularised LSTM across all six GJR-GARCH Monte Carlo stress scenarios. All results use the `StandardScaler` fitted exclusively on the training data (2010–2019), so the model receives in-

puts on the same scale as during training. Both raw directional accuracy and balanced accuracy are reported because the label distribution differs across scenarios. Balanced accuracy weights the true-positive and true-negative rates equally, so it is far less sensitive to the proportion of HIGH labels and better isolates genuine discriminative skill.

Scenario	Shock multiplier	Raw acc. (%)	Balanced acc. (%)
Normal	1.0×	55.37	56.38
COVID crisis magnitude	1.8×	56.85	57.58
2008 crisis magnitude	2.5×	56.43	56.88
Synthetic Extreme	3.0×	56.48	56.79
Structural Break	1.0× → 3.0× (day 180)	56.30	56.62
Non-linear Squeeze	Exponential (day 200)	56.48	55.61

Table 5: LSTM directional accuracy across the six stress scenarios. Raw and balanced accuracy are reported for each scenario.

Across all six scenarios, the model’s raw directional accuracy stays within a 1.5-percentage-point band, from 55.37% (Normal) to 56.85% (COVID crisis magnitude), and its balanced accuracy within a comparable band, from 55.61% to 57.58%. Accuracy does not decline as the simulated stress increases; every scenario sits between 0.93 and 1.48 percentage points above the Normal baseline in raw terms. The measured performance is therefore effectively flat across the full range of simulated stress, as illustrated in Appendix A.4. Section 4.4 tests whether these sub-two-point differences are statistically significant, and Section 4.5 examines whether they reflect a change in the model’s skill or a shift in the evaluation distribution.

4.4 Statistical Significance

To assess whether the accuracy differences observed in Table 5 are statistically significant, a two-sided proportion z-test was applied to each scenario against the Normal baseline, with $n = 10,000$ paths per scenario. Effect sizes are reported as Cohen’s h , and the results appear in Table 6.

Scenario	Δp	z	p -value	Sig.	Cohen's h
COVID crisis magnitude	+1.48	2.109	0.0350	*	0.030
2008 crisis magnitude	+1.06	1.510	0.1311	ns	0.021
Synthetic Extreme	+1.11	1.581	0.1139	ns	0.022
Structural Break	+0.93	1.324	0.1854	ns	0.019
Non-linear Squeeze	+1.11	1.581	0.1139	ns	0.022

Table 6: Proportion z-test results: each scenario against the Normal baseline ($n = 10,000$). Significance: * $p < 0.05$; ns = not significant. Bonferroni-corrected threshold: $\alpha = 0.01$.

The deltas relative to the Normal baseline are all of the order of one percentage point, and Cohen's h is negligible in every case ($h \leq 0.03$ throughout). Only the COVID crisis magnitude scenario reaches nominal significance ($z = 2.11$, $p = 0.035$), and even this does not survive the Bonferroni-corrected threshold ($\alpha = 0.01$) for five comparisons; the remaining scenarios are not significant. Because $n = 10,000$ paths per scenario gives the test high statistical power, even a one-percentage-point difference can reach nominal significance, which is why practical significance is judged from Cohen's h alongside the p -value. Taken together, the model's raw directional accuracy does not change materially across the stress scenarios; it neither collapses under severe stress nor improves in any practically meaningful way. This near-constant performance is itself a substantive result, examined further in Section 5.

4.5 Lookback Contamination Test

A conventional stress test evaluates a model at a single fixed window position and yields one accuracy figure. That figure alone cannot distinguish genuine robustness from a value that the evaluation scenario has inflated or depressed by shifting the proportion of HIGH-volatility labels in the window. To test whether the accuracy at any single position can be trusted as a measure of robustness, a systematic lookback contamination test varies the number of squeeze days visible within the 60-day feature window from 0 to 51, across 18 evenly spaced levels $c \in \{0, 3, 6, \dots, 51\}$.

Two findings emerge from Table 7. First, the HIGH-label proportion is far from constant across the 18 levels: it rises from roughly 43% at $c = 0$ to a peak of 71.8% at $c = 6$ and returns to 43% by $c = 51$. Against this swing, balanced accuracy stays within a narrow band, from 54.12% to 57.23% (a

spread of 3.11 percentage points, standard deviation 0.79), whereas raw accuracy moves over a wider range, from 52.23% to 56.87% (4.64 percentage points). The two extremes of squeeze visibility are statistically indistinguishable: a proportion z -test between the blind condition ($c = 0$, raw accuracy 56.30%) and the deepest-visibility condition ($c = 51$, raw accuracy 56.48%) returns $p = 0.80$.

Second, both accuracy measures are negatively correlated with the HIGH-label proportion, and the association is stronger for raw accuracy: $r = -0.85$ (95% CI $[-0.94, -0.63]$) against $r = -0.64$ (95% CI $[-0.85, -0.25]$) for balanced accuracy. Because the 18 levels are overlapping, non-independent windows, the difference between the two correlations was assessed with a path-level bootstrap ($B = 10,000$ resamples of the 10,000 paths), which places $r_{\text{raw}} - r_{\text{bal}}$ at -0.261 (95% CI $[-0.441, -0.108]$), excluding zero. Raw accuracy is therefore significantly more confounded with label composition than balanced accuracy. By averaging over the two classes, balanced accuracy strips out the prevalence effect that drives the raw measure, which is why it tracks the label distribution far less closely. Its residual correlation comes not from the label mix but from the per-class hit rates themselves, since squeeze depth shifts both the label share and the difficulty of the two classes at once. This sensitivity of both metrics under distributional shift is documented in the spurious-correlation literature (Ye et al., 2024) and formalised in the distribution-shift framework of Fan et al. (2023).

Contamination (days)	Raw acc. (%)	Balanced acc. (%)	HIGH label (%)
0	56.30	57.23	43.0
3	55.02	54.24	67.1
6	53.53	54.12	71.8
9	53.32	54.79	71.6
12	52.69	54.36	68.4
15	52.23	54.26	65.0
18	53.77	55.41	60.5
21	53.99	54.73	55.4
24	54.32	54.57	51.7
27	55.33	55.22	49.1
30	55.01	54.64	47.0
33	55.57	54.98	45.1
36	56.57	55.89	44.3
39	56.02	55.39	44.7
42	56.65	55.91	43.9
45	56.87	56.03	43.2
48	55.18	54.40	43.4
51	56.48	55.61	43.2

Table 7: Lookback contamination test: raw accuracy, balanced accuracy, and HIGH label proportion as a function of squeeze days visible in the 60-day feature window (18 contamination levels).

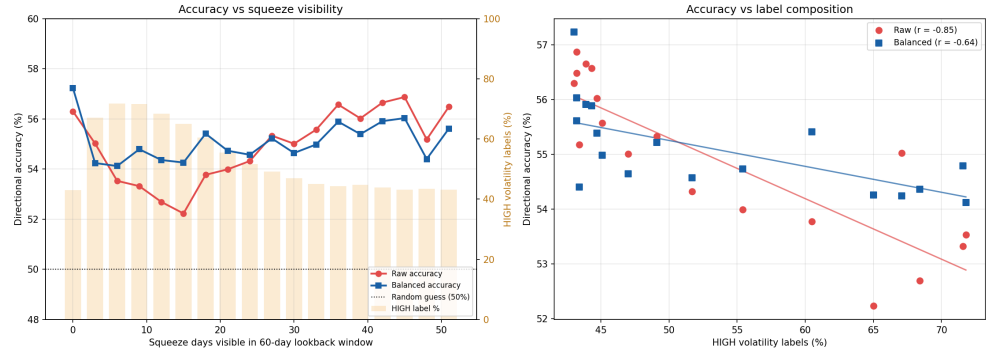


Figure 3: Lookback contamination test: raw accuracy tracks the HIGH-label proportion ($r = -0.85$) far more strongly than balanced accuracy ($r = -0.64$).

5 Discussion

5.1 Key Findings

The first sub-question concerned how the estimated GJR-GARCH parameters differ across the four historical market periods, an empirical finding from fitting the model to real S&P 500 data. The leverage parameter γ is positive in all four periods and significant in three, confirming that negative shocks raise future volatility more than positive shocks of equal size. The more consequential finding, however, is that 2008 and COVID-19 are not two severities of one crisis but two distinct regimes. The baseline variance ω rises roughly sevenfold from the pre-2008 to the COVID period, the GARCH coefficient β falls sharply, and the symmetric ARCH term α becomes non-zero only under COVID. The March 2020 shock arrived as an abrupt jump rather than a gradual build-up, so volatility became shock-driven rather than inertia-driven, which is what shifts weight from the GARCH coefficient β to the ARCH coefficient α . The 2008 crisis was endogenous, originating inside the financial system through leveraged credit, whereas COVID-19 was exogenous, a health shock with no financial precedent. The regimes differ not only in the magnitude of fear but also in the interpretability of its cause, and this difference registers directly in the parameters.

This distinction frames the second sub-question, which tested how LSTM performance changes under progressively scaled stress in magnitude. Raw directional accuracy remains within a narrow band of roughly 55–57%, with every scenario sitting marginally above rather than below the Normal baseline, and the effect sizes are negligible throughout (Cohen’s $h \leq 0.03$), with only the COVID scenario reaching even nominal significance, which does not survive Bonferroni correction. After the Bonferroni correction, no scenario differs significantly from the Normal baseline, and all gaps are under two percentage points. This near-constant performance is not interpreted as a null result to be explained away but as a direct consequence of how the stress scenarios are designed. Every scenario scales the pre-COVID variance ω while holding the dynamic parameters (α , γ , β) fixed, a *ceteris-paribus* experiment, so ω is the only parameter that varies across scenarios. Because the shock multiplier scales the baseline variance, a factor k corresponds to a $\sqrt{k}$ change in volatility, so the scenario labels denote relative stress intensity rather than the realised volatility of the crises they reference. The model is therefore always evaluated on the same regime it was trained on, only at

a higher magnitude; its consistent performance across scenarios is expected rather than evidence of robustness. The Structural Break holds α , β , and γ fixed and steps only ω at day 180, a mid-path change in level, not in persistence parameters, and therefore not an entirely structural change despite the name. The Non-linear Squeeze applies its multiplier as a deterministic overlay on σ_t after the GJR recursion, so the underlying data-generating process remains the frozen pre-COVID regime. This stress-testing framework was deliberately built within a single-axis boundary with varying magnitude, and that boundary is a design choice rather than an oversight.

The regularised LSTM (dropout and learning-rate scheduling) reaches an in-domain validation accuracy of 66.80% and balanced accuracy of 64.76%, a meaningful skill level on the calm 2010–2019 data. The balanced accuracy on the simulated Normal scenario (56.38%) sits well below the real held-out validation figure (64.76%), an eight-point gap between synthetic and real data drawn from the same pre-COVID process. This sim-to-real gap is a well-known limitation in machine learning, and it suggests that the GJR-GARCH Monte Carlo paths do not fully reproduce the structure the LSTM learned from real returns. As a result, accuracy settles at a simulation floor of roughly 55–57% across every synthetic scenario, irrespective of the stress applied, so the flat performance cannot be taken as evidence of the model’s robustness.

The third sub-question, the conceptual core of the thesis, examines the validity of the accuracy metric itself. The lookback contamination test, a moving-window evaluation, probes this directly by varying how much of the squeeze enters the model’s input and observing how each accuracy measure responds. As the window shifts and the squeeze enters the feature window, the proportion of HIGH labels rises from roughly 43% to 72% and returns to roughly 43%. This swing follows from the relative construction of the label where a day is classified HIGH only when its return exceeds the average of the preceding 20 days, so the reference level shifts as the window slides. Early in the squeeze this 20-day average is still drawn from calm days, so the elevated returns exceed it and the HIGH share rises to 72%. As the window advances, the average itself rises as it comes to include squeeze days; the same elevated returns no longer exceed it, and the share returns to 43%, since by the deepest level ($c = 51$) the squeeze has become the baseline against which each day is judged. Raw accuracy is strongly correlated with the HIGH-label proportion ($r = -0.85$) and balanced accuracy more weakly ($r = -0.64$). To assess this difference while accounting for the dependence between overlap-

ping windows, a path-level bootstrap resampling of the 10,000 paths places $r_{\text{raw}} - r_{\text{bal}}$ at -0.261 (95% CI $[-0.441, -0.108]$), confirming that raw accuracy is significantly more confounded with label composition than balanced accuracy. Raw accuracy weights each class by its share of the window, so $\text{Acc}(\pi) = \pi \text{TPR} + (1 - \pi) \text{TNR}$, with π the proportion of HIGH labels; its derivative is $\text{TPR} - \text{TNR}$ (Appendix A.5), so whenever the model is unequally accurate on the two classes, a shift in the label mix alters the raw figure by itself, even when the model is unchanged. The model classifies HIGH-volatility days less reliably than LOW-volatility days, which drives raw accuracy down as the HIGH-label proportion rises. Having the squeeze fully visible inside the feature window ($c = 51$) produces no higher accuracy than leaving it out of the window entirely ($c = 0$), the two conditions being statistically indistinguishable ($p = 0.80$). The model therefore draws little genuine predictive signal from the high-volatility structure, so its stable accuracy across the stress scenarios reflects limited skill and a label-driven metric rather than robustness.

5.2 Implications for Article 15 Compliance

Article 15 of the EU AI Act requires that high-risk systems achieve an appropriate level of accuracy, robustness, and cybersecurity, and perform consistently in those respects throughout their lifecycle. The central contribution of this thesis is to treat a model’s accuracy not as a standalone verdict but as a claim that must itself be validated before it can stand as the sole representation of robustness. A single accuracy figure, however carefully produced, compresses a great deal into one number, and the question this thesis presses is how much of that number can be trusted as a faithful measure of the model and how much reflects the conditions under which it was measured. This parallels an assessment of varying difficulty, in which an identical candidate attains a higher score on a less demanding paper, so the outcome reflects the composition of the items rather than the candidate’s underlying ability. The contamination test is a check that re-evaluates the model at a series of window positions with increasing amounts of the stressed period visible, and examines whether the accuracy figure tracks the model’s skill or the shifting label composition, before that figure is used to place the model in a risk band.

VaR analysis indicated to every major bank in 2007 that its risk was relatively low, estimating potential losses from a distribution of historical returns drawn almost entirely from calm periods, so the figure it produced was stable and consistent and passed every regulatory test (Jorion, 2009), until the

system collapsed in 2008. Regulators later responded with stressed VaR, scenario diversity, and capital surcharges, recognising that a metric which looks sound in normal conditions can be blind to periods of market stress, the very conditions a risk measure exists to capture (Basel Committee on Banking Supervision, 2009). For high-risk financial AI, establishing that a performance metric is valid before it is accepted as a final compliance figure could be a comparable prerequisite. The consequence is that robustness assessment under Article 15 is as much a governance question as a technical one. In the financial domain, the appropriate response is not to read the number at face value but to escalate human oversight, transferring the judgement to domain experts who can interpret the drivers of the stress and judge whether the metric still measures performance.

5.3 The Limits of Predictability: Relative versus Absolute Volatility

One of the major structural drawbacks of the stress-testing model is that it evaluates a crisis in relative terms, asking whether the market is moving sharply compared with its own recent history. What matters for risk management, however, is whether a crash will occur, which is an inherently absolute question and does not depend on how today compares with recent days. In calm markets the relative and absolute questions coincide, so the distinction goes unnoticed; in a crisis they diverge, because a relative label cannot track an absolute condition once the distribution shifts beneath it. The label is meant to capture financial stress, but it does not measure stress directly. It records only whether a day's return is large relative to its recent history and takes that as the sign of stress. This indirect route is unavoidable, because financial stress cannot be observed on its own and shows up only through its footprint in returns, namely unusually large price movements. While the market's statistical behaviour stays stable, a return that is large relative to its recent history is also a genuinely stressed one, so the label and true stress agree. Once the distribution shifts and the recent baseline itself rises, a genuine crash can resemble an ordinary market: the stressed day no longer stands out against its now-elevated neighbours, the label keeps measuring relative size, the model keeps predicting it, and relative size has come apart from the stress it was meant to represent.

This is the distinction that Narayanan and Kapoor (2024, pp. 42–44) iden-

tify within predictive AI: the forecasting of future outcomes from historical data, where some phenomena are genuinely predictable, and others resist prediction regardless of the model used. Predicting traffic congestion on a busy road improves reliably with more data, because its underlying drivers, such as road capacity, commuting patterns, and time of day, are stable, observable, and repeat with enough regularity for machine learning models to extract genuine signal. Predicting whether a specific individual will be involved in an accident at a specific moment is fundamentally different, because the outcome depends on qualitative, unobservable variables such as fatigue, distraction, or a child running into the road, that no historical dataset captures in that exact combination. Predicting next-day volatility in calm markets resembles the first, while anticipating a novel crisis from the return series alone resembles the second, which is why early-warning systems drawing on observable macro-financial drivers can capture signal that a return-only model cannot. A crisis is driven by economic, political, and behavioural forces that the return series itself does not record, so the label the model learns is only a shadow of a process that lies largely outside its data. This has been demonstrated empirically at scale: in a collaboration of more than 150 teams, rich data could predict individual life outcomes only marginally better than a simple baseline (Salganik et al., 2020, pp. 8398–8400), confirming that some outcomes resist prediction regardless of model sophistication.

Acknowledging these two problems together sharpens why the validity principle matters here in particular. The first is unavoidable, since crisis prediction depends on forces the return series never records, and no model fitted to historical or simulated data can fully capture them. The second is avoidable, since a single accuracy figure can mislead even when the model is sound. A systematic danger arises when a distorted metric is laid over a task that is already externally difficult, because the metric then conceals the error instead of exposing it and presents an unreliable result as a trustworthy one. Validating the metric does not make crisis prediction easy. It ensures that the genuine difficulty of predicting a crisis is not compounded by a metric that misstates the model’s true skill.

5.4 Limitations

This thesis contributes through its depth. A single index is carried through the full pipeline, from GJR-GARCH estimation to the lookback contamination test and its dependence-corrected bootstrap, at a depth that would be chal-

lenging to replicate across a broader set of assets. Several limitations follow from this scope.

Single asset. The analysis uses only the S&P 500. The contamination effect and parameter divergence may differ across other equities, fixed income, currencies, or commodities.

Simulated rather than realised paths. Scenarios are generated parametrically from GJR-GARCH estimates and do not capture liquidity freezes, contagion, or feedback loops present in a real crisis. The results are therefore based on synthetic paths whose realism is bounded by the estimated process.

Non-independent contamination levels. The eighteen levels are overlapping windows over the same paths, so the nominal Pearson p -values are anti-conservative. The path-level bootstrap in Section 4 addresses this, but inference still rests on a single simulated process.

Single model class and proxy label. The effect is demonstrated only for the LSTM, not transformer, GRU, or hybrid architectures, and only for the 20-day rolling-average label. Whether it generalises across model classes and label designs requires verification.

Oversight criteria not calibrated. The thesis proposes that human oversight escalates with measured degradation, yet the thresholds triggering escalation remain more principled than defined. Validating such thresholds across asset classes and crisis regimes is a precondition for regulatory use.

6 Conclusion

This thesis set out to examine whether an LSTM-based volatility prediction model remains robust under GJR-GARCH-generated stress scenarios relevant to Article 15 of the EU AI Act, and more fundamentally whether the accuracy metric used to evaluate that robustness stays valid under the conditions a stress test is built to create. Rather than surveying many assets, it takes a single index through the full pipeline, from GJR-GARCH estimation across four market regimes to Monte Carlo generation of six stress scenarios, an LSTM trained on calm 2010–2019 S&P 500 data and evaluated on 10,000 paths, and a lookback contamination test with a dependence-corrected bootstrap. The model’s accuracy stayed within a narrow band throughout, with no deviation surviving Bonferroni correction. This apparent stability is best understood as a property of the controlled design, because the structural persistence parameters (α, γ, β) were held fixed, and the model was evaluated only on a

higher-variance version of the regime it was trained on.

Under the shift induced by moving the feature window across the eighteen contamination levels, the label composition changed with the distribution of the squeezed-window data, and raw accuracy tracked that changing composition rather than the model’s skill. This carries a direct regulatory consequence, namely that a consistent number does not, on its own, validate the robustness of a model. Addressing it pushes the focus toward the procedural as well as the architectural, so that the validity of a metric is established before its value is trusted, and judgement escalates to human oversight wherever a metric proves confounded with the conditions of its own evaluation. A genuinely novel crisis is hard to forecast from historical returns, since a crisis is defined less by the magnitude of its price movements than by the specific constellation of causes behind them, and that exact pattern rarely repeats. This does not deny the early-warning-systems literature, which reports genuine predictive success from macro-financial and credit indicators such as credit growth and the slope of the yield curve (Bluwstein et al., 2023). Even so, each major crash carries its own distinct identity, whether the dot-com bubble, the 2008 financial crisis, or the COVID-19 shock, and a model trained on the dynamics of past crises may offer only a weak signal of the next, whereas a human reads a possible crisis by combining the data with outside signals that models lack. To conclude, this limitation must remain central to any crisis stress-testing framework, hence needs expert judgement and further empirical analysis before a standardised metric can reasonably be agreed upon. These findings are therefore best read as a proof of concept rather than an established regularity, resting on a single index and on parametrically simulated crisis paths, and bounded throughout by the limitations discussed above.

There are several directions that would deepen this work in the future. The most significant direction is to perturb the dynamics by varying the full parameter vector (α, γ, β) through a regime-switching data-generating process or jumps, so that the model is tested on structural change. A second is to repeat the analysis on larger LSTMs and on other model families used in financial forecasting, such as random forests, to examine whether the label-composition effect is a property of the metric. A third is to extend the framework to further asset classes, including equities, foreign exchange, and fixed income, to test whether the effect holds across different volatility structures.

References

- Basel Committee on Banking Supervision. Revisions to the Basel II market risk framework. Technical report, Bank for International Settlements, 2009. <https://www.bis.org/publ/bcbs158.pdf>.
- Fischer Black. Studies of stock market volatility changes. In *Proceedings of the 1976 Meetings of the American Statistical Association*, pages 177–181. American Statistical Association, 1976.
- Kristina Bluwstein, Marcus Buckmann, Andreas Joseph, Sujit Kapadia, and Özgür Şimşek. Credit growth, the yield curve and financial crisis prediction: Evidence from a machine learning approach. *Journal of International Economics*, 145, 2023. doi: 10.1016/j.jinteco.2023.103773.
- Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986. doi: 10.1016/0304-4076(86)90063-1.
- Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001. doi: 10.1023/A:1010933404324.
- Kay H. Brodersen, Cheng Soon Ong, Klaas E. Stephan, and Joachim M. Buhmann. The balanced accuracy and its posterior distribution. In *Proceedings of the 20th International Conference on Pattern Recognition (ICPR)*, pages 3121–3124, 2010. doi: 10.1109/ICPR.2010.764.
- Markus K. Brunnermeier. Deciphering the liquidity and credit crunch 2007–2008. *Journal of Economic Perspectives*, 23(1):77–100, 2009. doi: 10.1257/jep.23.1.77.
- Stephen Cecchetti, Robin L. Lumsdaine, Tuomas Peltonen, and Antonio Sánchez Serrano. Artificial intelligence and systemic risk. Technical Report 16, European Systemic Risk Board Advisory Scientific Committee, December 2025. https://www.esrb.europa.eu/pub/pdf/asc/esrb.ascreport202512_AIandsystemicrisk.en.pdf.
- Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, New York, 1993. ISBN 9780412042317.
- Robert F. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation. *Econometrica*, 50(4):987–1007, 1982. doi: 10.2307/1912773.

European Parliament and Council of the European Union. Regulation (EU) 2024/1689 — artificial intelligence act. Official Journal of the European Union, OJ L 2024/1689, July 2024. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32024R1689>.

Wei Fan, Pengyang Wang, Dongkun Wang, Dongjie Wang, Yuanchun Zhou, and Yanjie Fu. Dish-TS: A general paradigm for alleviating distribution shift in time series forecasting. In *Proceedings of the 37th AAAI Conference on Artificial Intelligence*, pages 7522–7529, 2023. doi: 10.1609/aaai.v37i6.25914.

Financial Stability Board. The financial stability implications of artificial intelligence. Technical report, Financial Stability Board, November 2024. <https://www.fsb.org/2024/11/the-financial-stability-implications-of-artificial-intelligence/>.

Thomas Fischer and Christopher Krauss. Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2):654–669, 2018. doi: 10.1016/j.ejor.2017.11.054.

Paul Glasserman. *Monte Carlo Methods in Financial Engineering*, volume 53 of *Stochastic Modelling and Applied Probability*. Springer, New York, 2003. ISBN 9780387004518.

Lawrence R. Glosten, Ravi Jagannathan, and David E. Runkle. On the relation between the expected value and the volatility of the nominal excess return on stocks. *The Journal of Finance*, 48(5):1779–1801, 1993. doi: 10.1111/j.1540-6261.1993.tb05128.x.

Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997. doi: 10.1162/neco.1997.9.8.1735.

Philippe Jorion. *Financial Risk Manager Handbook*. Wiley, 5th edition, 2009. ISBN 9780470479612.

Benoit Mandelbrot. The variation of certain speculative prices. *The Journal of Business*, 36(4):394–419, 1963.

Arvind Narayanan and Sayash Kapoor. *AI Snake Oil: What Artificial Intelligence Can Do, What It Can't, and How to Tell the Difference*. Princeton University Press, 2024. ISBN 9780691249131.

- Matthew J. Salganik et al. Measuring the predictability of life outcomes with a scientific mass collaboration. *Proceedings of the National Academy of Sciences*, 117(15):8398–8403, 2020. doi: 10.1073/pnas.1915006117.
- Ruey S. Tsay. *Analysis of Financial Time Series*. Wiley, Hoboken, NJ, 3rd edition, 2010. ISBN 9780470414354.
- Najlae Yachou, Omar Abahman, and Khalid Hakimi. Designing an LSTM-based model for financial asset forecasting using machine learning. *Central European Economic Journal*, 13(60):1–23, 2026. doi: 10.2478/ceej-2026-0001.
- Wenqian Ye, Guangtao Zheng, Xu Cao, Yunsheng Ma, and Aidong Zhang. Spurious correlations in machine learning: A survey. *arXiv preprint arXiv:2402.12715*, 2024. URL <https://arxiv.org/abs/2402.12715>.

Declaration of Authorship

I hereby declare that I have composed my Master’s thesis, “Stress Testing Automated Trading Systems using a Quantitative Framework for Financial Regulatory Systems” independently, using only the resources cited. All passages taken from publications, whether verbatim or in substance, have been identified as such. I agree that the work may be reviewed using plagiarism detection software. Neither this thesis nor any extract of it has been previously submitted to an examining authority in this or a similar form.

I confirm that the submitted written version is identical to the version saved on the enclosed storage medium. I have used AI technologies for purposes not included in the “Whitelist” published by the Department of Economics.

I used an AI-based language model (Claude) as a supporting tool during the preparation of this thesis. It was used for the following purposes:

Clarifying methodological concepts and literature (e.g., GARCH-family model specifications, LSTM model functionality, distributional shift, evaluation metrics).

Improving grammar, punctuation, and readability of the written text.

Technical support with the Python implementation, particularly debugging, error checking, and improving code structure and efficiency.

The research design, model implementation, GJR-GARCH estimation, Monte Carlo scenario generation, LSTM architecture and training, the look-back contamination protocol, the path-level bootstrap, the empirical analysis, and all interpretations of the results were carried out by the author. The author has independently verified all results, tables, figures, and references and takes full responsibility for the content of this thesis.

I have ensured that the written version of this thesis is identical to the version saved on the enclosed storage medium.

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A Appendix

This appendix collects the derivations, figures, and pseudocode referenced in the main text: the unconditional-variance derivation of the GJR-GARCH process (Section A.1), the stress-scenario path simulations and LSTM training curves (Sections A.2 and A.3), the stress-test accuracy chart (Section A.4), the prevalence dependence of accuracy (Section A.5), the methodology flowchart (Section A.6), and the lookback contamination algorithm (Section A.7).

A.1 Unconditional Variance of the GJR-GARCH Process

The stationarity condition and unconditional variance of the GARCH family are standard (Bollerslev, 1986); the derivation below follows the standard treatment for asymmetric GARCH models (Glosten et al., 1993; Tsay, 2010) and specialises it to the GJR-GARCH(1,1) case, where symmetric innovations yield the factor $\frac{1}{2}$ on γ .

The GJR-GARCH(1,1) process is

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma I_{t-1} \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (22)$$

where the shock is $\varepsilon_t = \sigma_t z_t$, the innovations z_t are independent and identically distributed with $\mathbb{E}[z_t] = 0$ and $\mathbb{E}[z_t^2] = 1$, and the indicator is $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$ and 0 otherwise. The goal is the unconditional variance $\bar{\sigma}^2 = \mathbb{E}[\sigma_t^2]$, the long-run average level.

Step 1: Take the unconditional expectation.

$$\mathbb{E}[\sigma_t^2] = \omega + \alpha \mathbb{E}[\varepsilon_{t-1}^2] + \gamma \mathbb{E}[I_{t-1} \varepsilon_{t-1}^2] + \beta \mathbb{E}[\sigma_{t-1}^2]. \quad (23)$$

Step 2: Relate $\mathbb{E}[\varepsilon_{t-1}^2]$ to $\mathbb{E}[\sigma_{t-1}^2]$. Since $\varepsilon_{t-1} = \sigma_{t-1} z_{t-1}$ with z_{t-1} independent of σ_{t-1} and $\mathbb{E}[z_{t-1}^2] = 1$,

$$\mathbb{E}[\varepsilon_{t-1}^2] = \mathbb{E}[\sigma_{t-1}^2] \mathbb{E}[z_{t-1}^2] = \mathbb{E}[\sigma_{t-1}^2]. \quad (24)$$

Step 3: Evaluate the asymmetry term. Because $\sigma_{t-1} > 0$, the sign of ε_{t-1} equals the sign of z_{t-1} , so $I_{t-1} = \mathbf{1}\{z_{t-1} < 0\}$. With symmetric innovations,

$$\mathbb{E}[I_{t-1} \varepsilon_{t-1}^2] = \mathbb{E}[\sigma_{t-1}^2] \mathbb{E}[z_{t-1}^2 \mathbf{1}\{z_{t-1} < 0\}] = \frac{1}{2} \mathbb{E}[\sigma_{t-1}^2], \quad (25)$$

since the negative half of a symmetric distribution contributes exactly half of $\mathbb{E}[z^2] = 1$. This produces the factor $\frac{1}{2}$ on γ .

Step 4: Apply stationarity. In the stationary state the variance level is constant over time,

$$\mathbb{E}[\sigma_t^2] = \mathbb{E}[\sigma_{t-1}^2] = \bar{\sigma}^2. \quad (26)$$

Step 5: Substitute Steps 2–4 into Step 1.

$$\bar{\sigma}^2 = \omega + \alpha \bar{\sigma}^2 + \gamma \cdot \frac{1}{2} \bar{\sigma}^2 + \beta \bar{\sigma}^2 = \omega + (\alpha + \beta + \frac{1}{2}\gamma) \bar{\sigma}^2. \quad (27)$$

Step 6: Solve for $\bar{\sigma}^2$. Collecting the $\bar{\sigma}^2$ terms gives

$$\bar{\sigma}^2 [1 - (\alpha + \beta + \frac{1}{2}\gamma)] = \omega \implies \bar{\sigma}^2 = \frac{\omega}{1 - (\alpha + \beta + \frac{1}{2}\gamma)}, \quad (28)$$

valid when $\alpha + \beta + \frac{1}{2}\gamma < 1$, the stationarity condition that keeps the denominator positive and the long-run variance finite.

Corollary: effect of the shock multiplier. Replacing ω by $k\omega$ while holding α , β , and γ fixed leaves the denominator unchanged, so

$$\bar{\sigma}_{\text{adj}}^2 = \frac{k\omega}{1 - (\alpha + \beta + \frac{1}{2}\gamma)} = k \bar{\sigma}^2, \quad \bar{\sigma}_{\text{adj}} = \sqrt{k} \bar{\sigma}. \quad (29)$$

The shock multiplier k therefore scales the long-run variance linearly and the long-run volatility by $\sqrt{k}$, confirming that it controls magnitude alone while the persistence, leverage asymmetry, and ARCH sensitivity remain those of the pre-COVID regime.

Long-run volatility of the stress scenarios. Applying the corollary above to the pre-COVID baseline ($\omega = 0.035730$, persistence $\alpha + \beta + \frac{1}{2}\gamma = 0.9558$) gives a baseline unconditional variance of $\bar{\sigma}^2 = 0.035730/(1 - 0.9558) \approx 0.808$, i.e. a long-run volatility of $\bar{\sigma} \approx 0.90\%$ per day. Scaling ω by k scales this level by $\sqrt{k}$:

Scenario	k	$\sqrt{k}$	Long-run vol. (%/day)
Normal	1.0	1.000	0.90
COVID crisis magnitude	1.8	1.342	1.21
2008 crisis magnitude	2.5	1.581	1.42
Synthetic Extreme	3.0	1.732	1.56

Table 8: Theoretical long-run (unconditional) volatility implied by each magnitude scenario, obtained by scaling the pre-COVID baseline ω by k . These are asymptotic levels of the data-generating process, distinct from the realised volatility of individual 252-day simulated paths. The Non-linear Squeeze is excluded because it multiplies σ_t directly (up to $M_{\max} = 5$) rather than scaling ω .

A.2 Stress Scenario Path Simulations

This appendix illustrates how the two time-varying scenarios evolve along a single simulated path. It plots the return series together with the conditional volatility it generates, so that the regime shift and the exponential squeeze can be seen directly rather than only through their summary statistics.

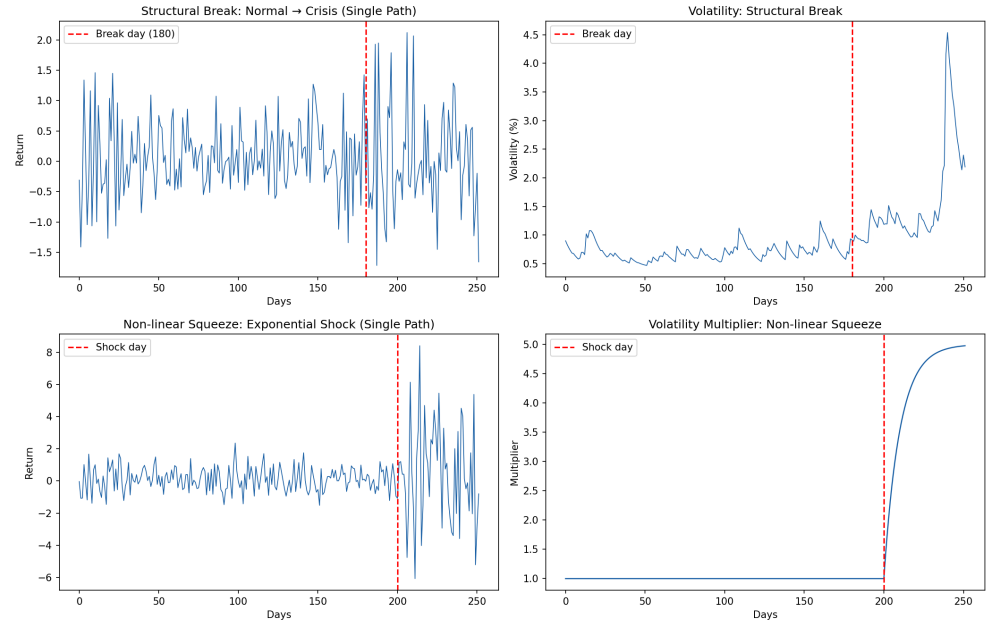


Figure 4: Representative single-path simulations. Top: Structural Break, returns (left) and conditional volatility (right); regime shift at day 180 (red dashed line). Bottom: Non-linear Squeeze, returns (left) and volatility multiplier $M(t)$ (right); shock onset at day 200, with $M(t) \rightarrow M_{\max} = 5.0$.

A.3 LSTM Training History

This appendix documents the convergence of the LSTM during training. It reports the training and validation loss and accuracy across epochs, showing where the validation loss stops improving and the point at which the best weights were restored (epoch 21).

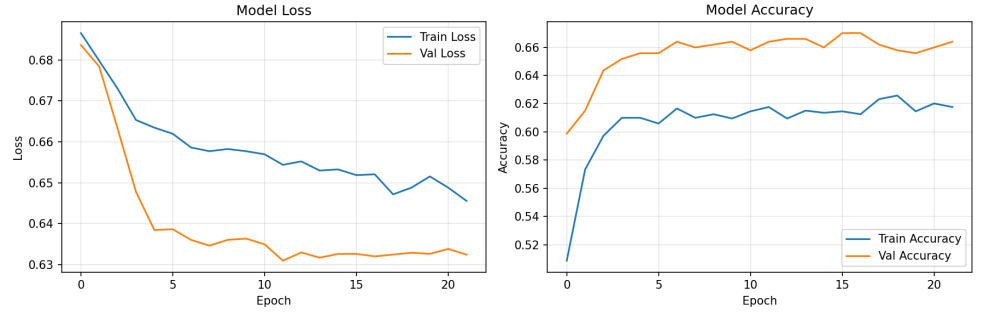


Figure 5: LSTM training and validation loss and accuracy over epochs. Best weights restored from epoch 21.

A.4 Stress Test Accuracy Chart

This appendix presents the directional accuracy of the LSTM across the six stress scenarios in graphical form. It complements the results table by placing each scenario against the 50% random-guess baseline and the Normal-scenario reference (55.37%).

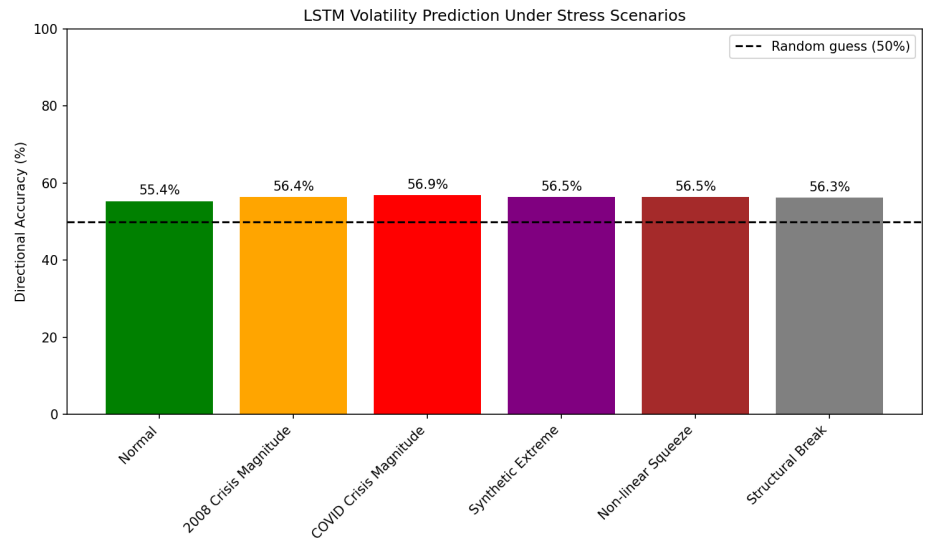


Figure 6: LSTM directional accuracy across the six stress scenarios. The dashed line marks the 50% random-guess baseline; the Normal scenario is the reference baseline (55.37%).

A.5 Prevalence Dependence of Accuracy

This appendix derives the relationship between raw accuracy, balanced accuracy, and the label prevalence π , which underlies the contamination analysis of Section 4.5. The derivation follows directly from the definitions of the two metrics.

Let $\pi = \pi_{\text{HIGH}}$ denote the proportion of HIGH-volatility labels in the evaluation set, and let the model's two class-conditional hit rates be

$$\text{TPR} = P(\hat{y} = 1 \mid y = 1), \quad \text{TNR} = P(\hat{y} = 0 \mid y = 0), \quad (30)$$

the true-positive rate (accuracy on HIGH days) and true-negative rate (accuracy on LOW days) respectively.

Raw accuracy. Raw accuracy is the probability of a correct classification, obtained by marginalising over the two classes weighted by their prevalence:

$$\text{Acc}(\pi) = \pi \text{TPR} + (1 - \pi) \text{TNR}. \quad (31)$$

Sensitivity to prevalence. Differentiating (31) with respect to π , holding the hit rates fixed, gives

$$\frac{\partial \text{Acc}}{\partial \pi} = \text{TPR} - \text{TNR}. \quad (32)$$

Raw accuracy therefore changes with the label prevalence whenever $\text{TPR} \neq \text{TNR}$, that is, whenever the model performs unequally on the two classes, even when its underlying discrimination is unchanged. The sign of the change is the sign of $\text{TPR} - \text{TNR}$ and carries no information about skill: a model stronger on LOW days ($\text{TNR} > \text{TPR}$) shows accuracy falling as π rises, while a model stronger on HIGH days shows the reverse.

Balanced accuracy. Balanced accuracy weights the two hit rates equally, independent of prevalence,

$$\text{BalAcc} = \frac{1}{2} \text{TPR} + \frac{1}{2} \text{TNR} \quad (33)$$

(Brodersen et al., 2010), so that

$$\frac{\partial \text{BalAcc}}{\partial \pi} = 0 \quad (34)$$

when the hit rates are held fixed. Balanced accuracy is thus independent of shifts in label prevalence and is equivalent to accuracy evaluated on a balanced test set ($\pi = 0.5$).

Numerical illustration. The consequence is visible directly. Suppose the model’s hit rates are fixed at $\text{TPR} = 0.60$ and $\text{TNR} = 0.50$, so that its underlying skill does not change. At a HIGH-label proportion of $\pi = 0.30$, raw accuracy is $0.30 \times 0.60 + 0.70 \times 0.50 = 0.53$; at $\pi = 0.70$ it becomes $0.70 \times 0.60 + 0.30 \times 0.50 = 0.57$. The reported figure therefore moves from 53% to 57% while the model itself is unchanged, purely because the label composition shifted. Balanced accuracy, fixed at $\frac{1}{2}(0.60 + 0.50) = 0.55$ for both values of π , does not move. This is the mechanism the contamination analysis isolates empirically.

Interpretation for the contamination test. Under the Non-linear Squeeze, the contamination protocol shifts π while the model itself is fixed. By (32), raw accuracy is dragged along with π , whereas balanced accuracy is, to first order, independent. The residual correlation observed for balanced accuracy ($r = -0.64$) reflects that the hit rates do not remain perfectly constant as the input window changes: the shift alters both the label prevalence and, more mildly, the model’s per-class performance. The difference between the two correlations therefore separates the component of the metric’s movement driven purely by label composition, which raw accuracy absorbs and balanced accuracy does not, from the component driven by a genuine change in discrimination, which both reflect.

A.6 Methodology Flowchart

The structure of the methodology flowchart is inspired by Yachou et al. (2026), extended with the stress-testing and lookback contamination stages introduced in this thesis. The pipeline runs from data collection and volatility labelling, through the train/validation split and the training of the regularised LSTM, to GJR-GARCH estimation and Monte Carlo scenario generation, and then to the stress-testing protocol and the lookback contamination test, whose outcomes are finally mapped to the risk-classification logic of Article 15.

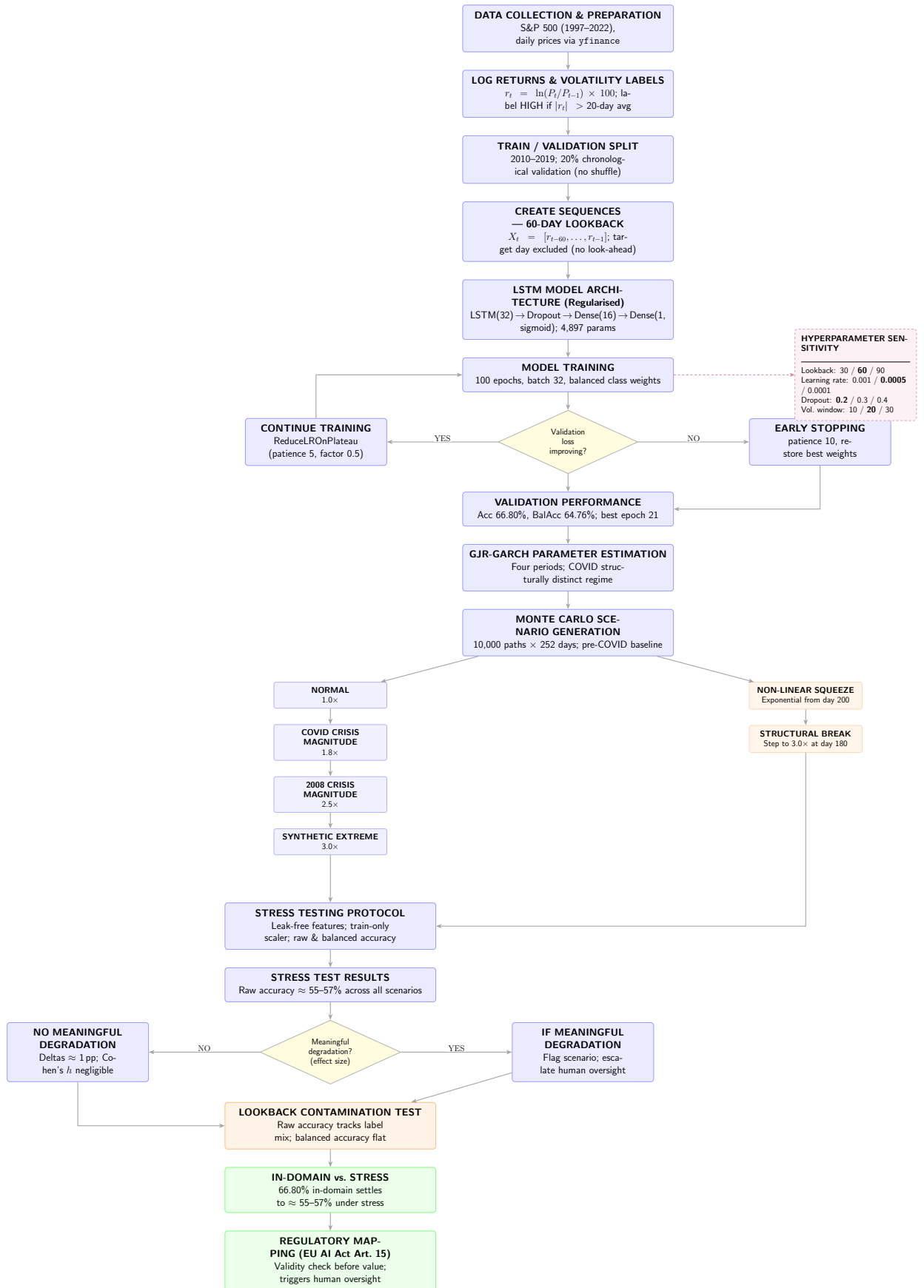


Figure 7: Methodology Flowchart for LSTM Volatility Prediction and Stress Testing.

A.7 Lookback Contamination Algorithm

Algorithm 1 summarises the lookback contamination protocol developed in this thesis. It combines two standard components, balanced accuracy (Brodersen et al., 2010) and a path-level bootstrap (Efron and Tibshirani, 1993), into a procedure for testing the validity of the accuracy metric under distributional shift.

Algorithm 1 Lookback Contamination Protocol

Require: Paths $\mathbf{P}$, model f_θ , scaler $(\hat{\mu}, \hat{\sigma})$, contamination set $\mathcal{C}$, shock onset $t^* = 200$, lookback $L = 60$, vol window $w = 20$

Ensure: Raw accuracy $\{\text{Acc}(c)\}$, balanced accuracy $\{\text{BalAcc}(c)\}$, label curve $\{\pi_{\text{HIGH}}(c)\}$, correlations $r_{\text{raw}}, r_{\text{bal}}$

```

1: for  $c \in \mathcal{C}$  do
2:    $t_{\text{end}} \leftarrow t^* + c$ 
3:    $t_{\text{start}} \leftarrow t_{\text{end}} - L$ 
4:    $\mathbf{X}(c) \leftarrow \mathbf{P}[:, t_{\text{start}} : t_{\text{end}}]$ 
5:    $\tilde{\mathbf{X}}(c) \leftarrow (\mathbf{X}(c) - \hat{\mu}) / \hat{\sigma}$ 
6:    $\mathbf{y}(c) \leftarrow \text{VolatilityLabel}(\mathbf{P}, t_{\text{end}}, w)$ 
7:    $\hat{\mathbf{y}}(c) \leftarrow \mathbf{1}(f_\theta(\tilde{\mathbf{X}}(c)) > 0.5)$ 
8:    $\text{Acc}(c) \leftarrow \text{mean}(\hat{\mathbf{y}}(c) = \mathbf{y}(c))$ 
9:    $\text{BalAcc}(c) \leftarrow \text{BalancedAccuracy}(\mathbf{y}(c), \hat{\mathbf{y}}(c))$ 
10:   $\pi_{\text{HIGH}}(c) \leftarrow \text{mean}(\mathbf{y}(c))$ 
11: end for
12:  $r_{\text{raw}} \leftarrow \text{Corr}(\pi_{\text{HIGH}}, \text{Acc})$ 
13:  $r_{\text{bal}} \leftarrow \text{Corr}(\pi_{\text{HIGH}}, \text{BalAcc})$ 
14: return  $\{\text{Acc}(c)\}, \{\text{BalAcc}(c)\}, \{\pi_{\text{HIGH}}(c)\}, r_{\text{raw}}, r_{\text{bal}}$ 

```
